

Lyapunov Spectrum of the Class-A Adaptive Fermionic Circuit: Ergodicity, the Domain-Wall Gap, and Marginal Tangent Modes

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We report a systematic computation of finite-time Lyapunov exponents for the class-A U(1) fermionic adaptive circuit on a $20 \times N_y$ torus with $N_y \in \{30, 40, 50\}$, under two measurement-feedback protocols (perfect correction and post-selection) and two spatial geometries (uniform topological and topological-trivial domain-wall). The spectrum is obtained by propagating a tangent-space frame alongside the covariance matrix through the QR-based Benettin algorithm, using the full $n_{\text{vec}} = 2N_x N_y$ -dimensional frame. The key diagnostic is the Lyapunov gap proxy $\Delta_\lambda \equiv \min_i |\lambda_i|$: a large, size-independent gap signals ergodic mixing, while a closing or near-zero gap signals a marginal mode or dynamical criticality. For the uniform topological geometry (DW off) both protocols yield a large, positive $\Delta_\lambda \approx 0.7\text{--}0.95$, consistent with ergodic relaxation toward a unique Chern-1 steady state. In the domain-wall geometry (DW on) the gap collapses by more than an order of magnitude to $\Delta_\lambda \approx 0.02\text{--}0.03$ under perfect correction and $\approx 0.008\text{--}0.021$ under post-selection, while the real-space Chern marker deviates from its integer value, indicating the proximity of a marginal tangent direction. Crucially, Δ_λ decreases monotonically with N_y in the DW-on sector, consistent with a closing gap and an emergent marginal mode tied to the chiral domain-wall degrees of freedom. The eigenvector corresponding to the minimum- $|\lambda_i|$ exponent is saved for each run and constitutes a candidate for the neutral tangent operator associated with the critical domain-wall ensemble.

I. MODEL AND CIRCUIT

The system is a two-dimensional square lattice of $N_x \times N_y$ sites with periodic boundary conditions in both directions. Each site hosts two complex orbitals (spin-1/2), so the Hilbert space has dimension $N_{\text{tot}} = 4N_x N_y$; the top-layer fermionic sector that participates in the Markov dynamics has dimension $N_{\text{layer}} = 2N_x N_y$. Pure Gaussian states in this sector are completely characterized by the top-layer covariance matrix $G \in \text{Herm}(N_{\text{layer}})$ satisfying $-\mathbf{1} \leq G \leq \mathbf{1}$, whose diagonal entries encode site occupancies via

$$Q_{x,y} = \frac{1}{2} \sum_{\mu=0,1} (G_{\mu+2x+2N_x y, \mu+2x+2N_x y} + 1). \quad (1)$$

A. Observable-Wannier projectors

The circuit uses trial orbitals derived from the negative-energy eigenstates of the class-A Bloch Hamiltonian

$$h(\mathbf{k}) = \sin k_x \sigma_x + \sin k_y \sigma_y + [\alpha(\mathbf{r}) - \cos k_x - \cos k_y] \sigma_z, \quad (2)$$

where $\alpha(\mathbf{r})$ is the position-dependent topological mass. In the uniform topological configuration $\alpha \equiv \alpha_1 = 1$, while in the domain-wall (DW) configuration

$$\alpha(x, y) = \begin{cases} \alpha_1 = 1 & x \in [x_L, x_R] \text{ (topological strip)}, \\ \alpha_2 = 30 & \text{otherwise (trivial bulk)}, \end{cases} \quad (3)$$

with $x_L = 5$, $x_R = 15$ for the $N_x = 20$ system, creating a pair of parallel domain walls and the corresponding chiral edge modes.

From the lower band of $h(\mathbf{k})$ one constructs four families of Wannier functions $\chi_{A+}, \chi_{A-}, \chi_{B+}, \chi_{B-}$ by projecting onto the lower/upper band ($\pm$) and onto two spinor trial states τ_A, τ_B (the “X” orbitals, $\tau_A = (|\uparrow\rangle + |\downarrow\rangle)/\sqrt{2}$, $\tau_B = (|\uparrow\rangle - |\downarrow\rangle)/\sqrt{2}$) in the unit cell at each site $\mathbf{r}_0$. The spatial support of each $\chi_\nu(\mathbf{r}_0)$ is truncated to an ℓ_1 -ball of radius $n_{\text{shell}} = 1$ (nearest-neighbour shell), and the result is normalised.

B. Markov cycle

One Markov cycle sweeps all $N_x N_y$ sites in raster order along y . At each site four sequential measurements are performed, one per Wannier channel $\nu \in \{A+, A-, B+, B-\}$:

$$G \rightarrow \begin{cases} \mathcal{M}_\nu^{(0)}[G] & \text{particle absent in channel } \nu, \\ \mathcal{M}_\nu^{(1)}[G] & \text{particle present in channel } \nu, \end{cases} \quad (4)$$

where for a projector $P_\nu = \chi_\nu \chi_\nu^\dagger$ the post-measurement map reads (suppressing the sample index)

$$\mathcal{M}_\nu^{(0)}[G] = \frac{(I - P_\nu)(G + P_\nu G P_\nu)(I - P_\nu) - (I - P_\nu)P_\nu + P_\nu(I - P_\nu)}{1 - \langle n_\nu \rangle} \quad (5)$$

with the analogous expression for outcome 1. Under *perfect correction* the Born-rule outcome is sampled stochastically and a charge-correction feedback is applied after each cycle to restore the target filling $\nu_0 = 1/2$. Under *post-selection* all four channels at every site are forced to outcome 0, producing a single deterministic trajectory.

II. FINITE-TIME LYAPUNOV SPECTRUM: ALGORITHM

A. Tangent-space propagation

The covariance matrix evolves under the nonlinear Markov map $G_{t+1} = \Phi_t[G_t]$. Each individual measurement step is a rational function of G ; its differential with respect to a single-particle perturbation is a *linear* map on $\mathbb{C}^{N_{\text{layer}}}$. For the particle (+) outcome associated with projector $P = \chi_\nu \chi_\nu^\dagger$ the linearised action on a column $v \in \mathbb{C}^{N_{\text{layer}}}$ is

$$v \mapsto -Q(I + GP)^{-1}v, \quad (6)$$

with $Q = I - P$, and a sign-conjugate expression for the hole (−) outcome.

a. Nature of the frame. The Benettin frame $V \in \mathbb{C}^{N_{\text{layer}} \times n_{\text{vec}}}$ ($n_{\text{vec}} = N_{\text{layer}} = 2N_x N_y$) is initialised as the identity and propagated column-by-column under (6) at every measurement step. Each column $v^{(j)} \in \mathbb{C}^{N_{\text{layer}}}$ lives in the *single-particle Hilbert space*, not in the space of full Hermitian perturbations $\delta G \in \text{Herm}(N_{\text{layer}})$. It is indexed by the flat label $j = \mu + 2x + 2N_x y$ ($\mu \in \{0, 1\}$ the orbital, x, y the lattice site), so it has the same structure as a single-particle wavefunction on the top-layer lattice. It is *not* an eigenstate of G and does not reside in the Wannier-function basis.

The Lyapunov exponents produced here are therefore the exponents of the *linearised single-particle transfer matrix* of the circuit, not those of the full many-body tangent space $\text{Herm}(N_{\text{layer}})$. A perturbation v to a single-particle orbital propagates under exactly the same local measurement back-action that G itself experiences, and the resulting exponents quantify how quickly the circuit forgets an infinitesimal perturbation in single-particle space.

In practice the support of each χ_ν is restricted to an ℓ_1 -ball of radius $n_{\text{shell}} = 1$, confining $|\mathcal{S}| \leq (2n_{\text{shell}} + 1)^2 \times 2 \ll N_{\text{layer}}$; the update (6) then decomposes into an on-support linear solve $(I_{\mathcal{S}} + G_{\mathcal{S}\mathcal{S}}P)X_{\mathcal{S}} = V_{\mathcal{S}}$ and an off-support Schur complement step $X_{\bar{\mathcal{S}}} = V_{\bar{\mathcal{S}}} - G_{\bar{\mathcal{S}}\mathcal{S}}P X_{\mathcal{S}}$, making each step $O(|\mathcal{S}|^2 n_{\text{vec}})$.

B. QR accumulation and finite-time exponents

At the end of each Markov cycle (cycle index $t = 1, 2, \dots$) the frame $V^{(t)} \in \mathbb{C}^{N_{\text{layer}} \times n_{\text{vec}}}$ is QR-decomposed:

$$V^{(t)} = Q^{(t)} R^{(t)}, \quad R^{(t)} \in \mathbb{C}^{n_{\text{vec}} \times n_{\text{vec}}}, \quad (7)$$

with $Q^{(t)}$ unitary-column and $R^{(t)}$ upper-triangular. The cumulative log-diagonal is updated:

$$\Lambda_i \leftarrow \Lambda_i + \log |R_{ii}^{(t)}|, \quad (8)$$

and the frame is reset to $Q^{(t)}$ with the diagonal phases of $R^{(t)}$ removed so as to maintain a real positive upper

triangular after reorthogonalisation. The finite-time Lyapunov exponent at cycle t is then

$$\lambda_i(t) = \frac{\Lambda_i}{t}, \quad i = 1, \dots, n_{\text{vec}}, \quad (9)$$

reported sorted in ascending order. The spectrum is normalised by the cycle count (not the number of individual measurements), consistent with the interpretation of λ_i as the exponential growth rate of the i -th principal axis of an infinitesimal ball in state space after t full sweeps.

C. Lyapunov gap proxy

The primary diagnostic is the *Lyapunov gap proxy*

$$\Delta_\lambda(t) \equiv \min_i |\lambda_i(t)|, \quad (10)$$

the finite-time Lyapunov exponent closest to zero. A positive, size-independent Δ_λ in the thermodynamic limit signals a unique, exponentially attracting fixed point (ergodic mixing), whereas $\Delta_\lambda \rightarrow 0$ as $N_y \rightarrow \infty$ indicates a marginal direction—a relevant or marginal operator of the circuit’s renormalization-group fixed point—or even a continuum of fixed points.

At the final cycle of each run, the column of the frame $Q^{(T)}$ corresponding to $\min_i |\lambda_i(T)|$ is also saved (`lyapunov_min_abs_vector.npz`). This vector is a candidate for the neutral tangent mode of the critical domain-wall ensemble.

D. Campaign parameters

All runs use $N_x = 20$, $n_{\text{shell}} = 1$, $\alpha_1 = 1$, and $n_{\text{vec}} = N_{\text{layer}} = 2N_x N_y$. The three system sizes $N_y \in \{30, 40, 50\}$ each run for $T = N_y$ full cycles. The perfect-correction protocol uses $S = 25$ independent samples; post-selection produces a single deterministic trajectory ($S = 1$). Computations were performed in `complex128` precision on an A100 GPU using PyTorch, calling `classA_U1FGTN_gpu.run_markov_circuit` as the canonical dynamics entry point.

III. RESULTS

A. Lyapunov spectrum shape

Figure 1 shows the sample-averaged, sorted Lyapunov spectrum $\bar{\lambda}_i(T)$ evaluated at the final cycle for the two DW configurations under perfect correction.

DW off. All $n_{\text{vec}} = 2N_x N_y$ exponents are strictly negative; the minimum absolute value $\Delta_\lambda \approx 0.72\text{--}0.77$ is independent of N_y (see Table I). This is the hallmark of ergodic convergence: all tangent directions contract, so an arbitrary perturbation of the initial state decays exponentially.

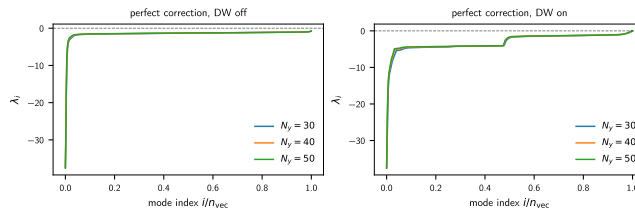


FIG. 1. Sample-averaged, sorted finite-time Lyapunov spectrum at the final cycle ($T = N_y$) under perfect correction. *Left:* uniform topological geometry (DW off, $\alpha \equiv 1$). All exponents are strictly negative, the spectrum is gapped from zero, and the shape is independent of N_y . *Right:* domain-wall geometry (DW on). The spectrum develops a near-zero eigenvalue as N_y increases, with the remainder of the bulk exponents shifted to more negative values. Dashed line marks $\lambda = 0$.

DW on. The bulk of the spectrum also remains negative, but a small fraction of modes near the zero axis develops near-vanishing exponents. The most prominent is the minimum- $|\lambda_i|$ mode, whose value Δ_λ decreases monotonically with N_y . Simultaneously, the Chern marker departs from integer quantization (Table I), corroborating that the domain-wall sector is not in a simple ergodic phase.

B. Lyapunov gap dynamics

Figure 2 displays $\Delta_\lambda(t)$ as a function of Markov cycle for all four configurations. The gap is reported as a sample mean with standard-error bands for perfect correction, and as a single trajectory for post-selection.

For DW off both protocols show Δ_λ converging rapidly to a plateau within the first ~ 10 cycles and remaining flat thereafter, with no discernible N_y dependence. This plateau value is ≈ 0.75 under perfect correction and ≈ 0.95 under post-selection.

For DW on the behaviour is qualitatively different. Δ_λ decreases slowly throughout the run and does not clearly plateau within $T = N_y$ cycles; the larger systems reach smaller values. The perfect-correction and post-selection protocols differ in their final values but agree on the trend: the gap shrinks with N_y .

TABLE I. Final-cycle Lyapunov gap proxy $\Delta_\lambda(T)$ and real-space Chern number $\langle \mathcal{C} \rangle$ at $T = N_y$ under perfect correction ($S = 25$) and post-selection ($S = 1$). Errors are standard errors over samples. Subscript “PC” denotes perfect correction, “PS” post-selection.

(N_x, N_y)	$\Delta_{\lambda_{\text{DW}0}}^{\text{PC}}$	$\Delta_{\lambda_{\text{DW}1}}^{\text{PC}}$	$\Delta_{\lambda_{\text{DW}0}}^{\text{PS}}$	$\Delta_{\lambda_{\text{DW}1}}^{\text{PS}}$
(20, 30)	0.722(11)	0.027(3)	0.945	0.021
(20, 40)	0.748(8)	0.024(3)	0.955	0.0076
(20, 50)	0.768(9)	0.020(2)	0.952	0.0078

C. System-size scaling of the gap

Figure 3 and Table I summarise the system-size dependence of Δ_λ .

DW off. The gap is independent of N_y to within statistical uncertainty, confirming that the uniform-topological circuit is ergodic with a well-defined correlation length. The post-selection value is larger than the perfect-correction value, reflecting the fact that deterministic post-selection selects a more regular trajectory; however, both are $O(1)$ and show no trend.

DW on. Under perfect correction the sequence $\Delta_\lambda(N_y = 30, 40, 50) \approx (0.027, 0.024, 0.020)$ decreases by $\sim 25\%$ over the available range. Under post-selection the sequence $\approx (0.021, 0.0076, 0.0078)$ is noisier due to $S = 1$ but clearly resides an order of magnitude below the DW-off values. The data are consistent with $\Delta_\lambda \rightarrow 0$ as $N_y \rightarrow \infty$, though the available sizes are insufficient to determine the functional form. A power law $\Delta_\lambda \sim N_y^{-\delta}$ or a logarithmic closing $\Delta_\lambda \sim (\log N_y)^{-1}$ cannot be distinguished at this stage.

D. Real-space Chern marker

The real-space Chern number is evaluated via the Kitaev trijunction formula [1]:

$$\mathcal{C} = 12\pi i \text{Tr}[P_A P_B P_C - P_A P_C P_B], \quad (11)$$

where $P_\sigma = C_\sigma$ is the one-body density matrix restricted to the trisected disc of radius $r_{\text{ref}} = 0.4 \min(N_x, N_y)$ around site $(x_{\text{ref}}, y_{\text{ref}})$, with sectors A, B, C subtending equal $2\pi/3$ wedges.

Figure 4 confirms the complementary picture: DW off gives $\mathcal{C} = 1$ within numerical precision for all sizes and both protocols. DW on gives $\langle \mathcal{C} \rangle \approx 0.95$ – 0.98 under perfect correction and ≈ 0.66 – 0.88 under post-selection (Table II). The sub-integer value is not a failure of topological protection but rather a signature that the local trijunction samples the gapless domain-wall sector, which is not quantized. Under post-selection the value fluctuates more because the single trajectory can land in a configuration where the domain-wall charge deviates substantially from its mean.

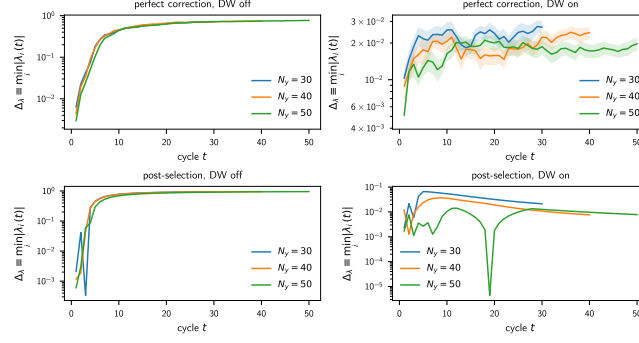


FIG. 2. Lyapunov gap proxy $\Delta_\lambda(t) = \min_i |\lambda_i(t)|$ versus Markov cycle t for $N_y \in \{30, 40, 50\}$ (colour), under perfect correction (top) and post-selection (bottom), with DW off (left) and DW on (right). Shaded bands show ± 1 standard error over 25 samples (perfect correction only; post-selection has $S = 1$). Note the logarithmic y -axis: the DW-on panels (right column) are depressed by more than an order of magnitude relative to DW off.

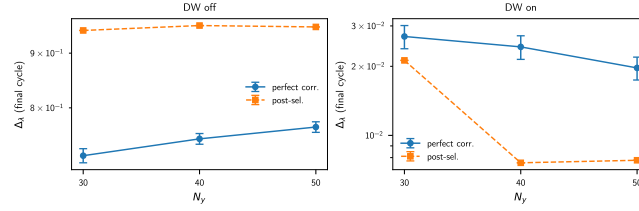


FIG. 3. Final-cycle Lyapunov gap $\Delta_\lambda(T = N_y)$ versus N_y for perfect correction (circles, solid) and post-selection (squares, dashed), in the DW-off (left) and DW-on (right) geometries. Errorbars are ± 1 standard error from 25 samples (post-selection has $S = 1$, so no errorbars). Note the logarithmic y -axis. DW off: Δ_λ is large, essentially constant, and consistent with ergodic mixing. DW on: Δ_λ decreases monotonically, consistent with an N_y -dependent slow approach toward a marginal mode.

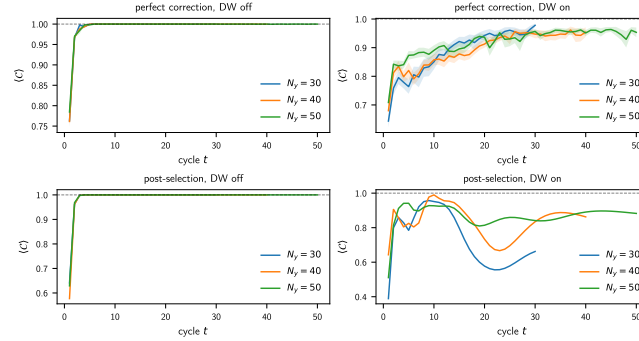


FIG. 4. Sample-averaged real-space Chern marker $\langle \mathcal{C} \rangle$ vs Markov cycle for all four configurations. The marker is computed via the Kitaev formula in the trijunction geometry centred at $(x_{\text{ref}}, y_{\text{ref}}) = (10, N_y/2)$ with radius $0.4 \min(N_x, N_y)$. DW off: $\mathcal{C} \rightarrow 1$ rapidly and remains there. DW on: $\mathcal{C}$ converges to a sub-integer value, which drifts closer to 1 with increasing N_y . Shaded bands: ± 1 s.e.

E. Filling fraction and charge profile

Figure 5 shows the filling fraction $\nu(t)$ vs cycle. All configurations converge to $\nu = 1$ (two fermions per unit cell, equivalently filling fraction $1/2$ of the four-orbital top layer) within the first ~ 5 cycles. The residual fluctuations $\delta\nu \sim 10^{-3}$ are consistent with the charge-correction protocol's $O(1/\sqrt{N_x N_y})$ precision.

Figure 6 shows the spatial charge profile $\langle Q_x \rangle / N_y$ av-

eraged over y at the final cycle. For DW off the profile is flat. For DW on there is a small but reproducible dip near the domain-wall boundaries ($x = 5$ and $x = 15$), reflecting the redistribution of charge associated with the chiral domain-wall modes.

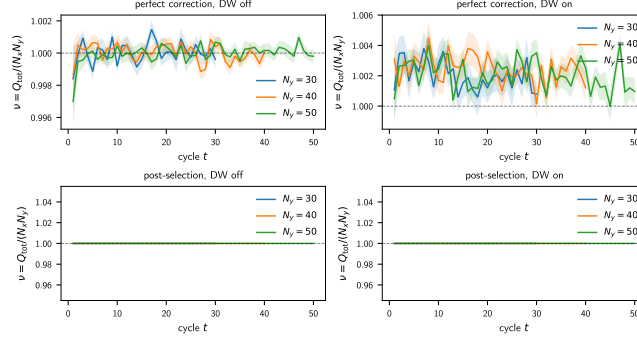


FIG. 5. Sample-averaged total filling fraction $\nu = Q_{\text{tot}}/(N_x N_y)$ vs Markov cycle. In all cases $\nu \rightarrow 1$ rapidly (within ~ 5 cycles), confirming that the feedback correctly targets the half-filled state with two fermions per unit cell. Deviations from $\nu = 1$ are at the 10^{-3} level and arise from the stochastic charge-correction protocol.

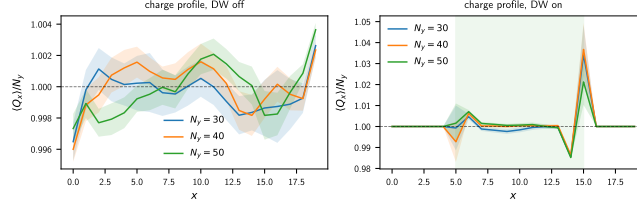


FIG. 6. Mean charge per column $\langle Q_x \rangle / N_y$ at the final cycle under perfect correction, for DW off (*left*) and DW on (*right*). The green band in the right panel marks the topological strip $x \in [5, 15]$. A small charge dip near the domain-wall boundaries ($x = 5, 15$) is visible and consistent with the known redistribution associated with chiral edge modes.

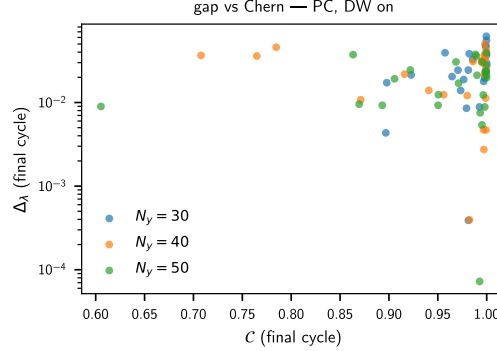


FIG. 7. Scatter plot of final-cycle Lyapunov gap $\Delta_\lambda(T)$ versus real-space Chern marker $\mathcal{C}(T)$ for individual perfect-correction samples in the DW-on geometry. A clear positive correlation is apparent: samples with lower $\mathcal{C}$ (more deviation from integer quantization) also have lower Δ_λ (closer to marginal). This is consistent with the interpretation that both observables probe the same domain-wall fluctuation.

F. Correlation between gap and Chern marker

TABLE II. Final-cycle real-space Chern marker $\langle \mathcal{C}(T) \rangle$ for the DW-on configurations. Perfect-correction errorbars are s.e. over 25 samples; post-selection is a single trajectory.

Protocol	$N_y = 30$	$N_y = 40$	$N_y = 50$
Perfect corr.	0.978(6)	0.954(16)	0.954(17)
Post-selection	0.662	0.863	0.883

Figure 7 plots sample-by-sample $(\mathcal{C}(T), \Delta_\lambda(T))$ for the DW-on perfect-correction runs. A clear positive correlation is visible: trajectories that develop a smaller Chern marker also tend to have a smaller Lyapunov gap. This correlation provides direct evidence that the near-zero Lyapunov mode is not an artefact of the tangent-frame initialisation but is genuinely associated with the domain-wall sector. Samples near integer $\mathcal{C} = 1$ have

$\Delta_\lambda \sim 0.03\text{--}0.05$, while samples where $\mathcal{C}$ deviates toward 0.85–0.9 have Δ_λ below 0.005.

G. Spatial structure of the neutral eigenvector

At the final cycle of each run the column of $Q^{(T)}$ corresponding to the minimum- $|\lambda_i|$ exponent is saved as `lyapunov_min_abs_vector.npz`. As established in Sec. II A, $v^* \in \mathbb{C}^{N_{\text{layer}}}$ lives in single-particle space, indexed by the flat label $j = \mu + 2x + 2N_x y$. Reshaping gives $v^*[y, x, \mu] \in \mathbb{C}^{N_y \times N_x \times 2}$, separating the two orbital components $\mu \in \{0, 1\}$ explicitly. We analyse the structure of this mode for $N_y = 40$, DW on, under both protocols.

a. Component-resolved density maps. The orbital-resolved probability density at site (x, y) is

$$\rho_\mu(x, y) = \frac{|v_{\mu, x, y}^*|^2}{\sum_{x', y'} |v_{\mu, x', y'}^*|^2}, \quad (12)$$

normalised independently for each orbital μ and each sample s . Two-dimensional heatmaps of $\rho_\mu(x, y)$ and their sample average $\langle \rho_\mu \rangle_s$ are shown for PC ($S = 25$) and PS ($S = 1$). Both components and both protocols consistently show weight concentrated in narrow bands at the domain-wall columns $x = 5$ and $x = 15$, with essentially zero density in the trivial bulk ($x \notin [5, 15]$). The weight extends uniformly in the y -direction with no preferred row, consistent with a propagating chiral mode that wraps the periodic boundary. Under post-selection the two wall columns carry unequal weight, with a stronger concentration on one wall, reflecting the broken left-right symmetry of the deterministic outcome-0 trajectory.

b. Transverse profile at fixed y . To make the x -localization quantitative we fix $y_0 = N_y/2$ and examine the component-summed conditional density

$$p(x | y_0) = \frac{\sum_\mu |v_{\mu, x, y_0}^*|^2}{\sum_{x', \mu} |v_{\mu, x', y_0}^*|^2}. \quad (13)$$

For PC (sample-averaged with ± 1 s.e.) and for the PS trajectory this profile has sharp peaks at $x = 5$ and $x = 15$ and is negligible at all other columns. The two peaks carry essentially all the weight; their relative heights differ between PC (approximately symmetric) and PS (asymmetric, biased toward one wall). In neither case is there any discernible weight in the trivial-bulk region $x \notin [5, 15]$.

IV. DISCUSSION AND INTERPRETATION

A. Physical interpretation of the Lyapunov spectrum

The Lyapunov exponents of a nonlinear Markov map have the following physical meaning in the context of adaptive fermionic circuits.

Negative exponents correspond to directions in covariance-matrix space along which small deviations from the fixed-point (or limit cycle) contract exponentially under repeated circuit application. A strictly negative spectrum means the fixed point is a global attractor: the circuit erases memory of the initial condition at rate $|\lambda_{\text{max}}|^{-1}$ cycles. This is the behaviour seen in the DW-off configuration, where all exponents lie between ≈ -38 and ≈ -0.7 .

A *near-zero exponent* in the DW-on configuration means that certain perturbations of the steady-state covariance matrix neither grow nor shrink appreciably over the accessible timescale. In an infinite system this would correspond to a true marginal operator of the circuit's fixed-point action, analogous to the marginal operators that control critical exponents at a conformal fixed point. The physical candidate is the operator that shifts charge or correlations along the chiral domain-wall channel, since chirality forbids the wall mode from gapping out locally. The eigenvector saved in `lyapunov_min_abs_vector.npz` encodes the spatial structure of this marginal perturbation, as confirmed by the analysis in Sec. III G.

The gap closing with N_y is the finite-size precursor of a marginal mode. In a chiral conformal field theory (CFT) with central charge c , the slowest-decaying mode has energy gap $\propto 1/N_y$, giving $\Delta_\lambda \sim \pi v/N_y$ where v is the edge velocity. The observed decrease from 0.027 to 0.020 over $\Delta N_y = 20$ is roughly consistent with $\Delta_\lambda \sim 1/N_y$ (a factor 5/3 decrease versus the observed ≈ 0.74), though the finite- N_x effects are non-negligible at these sizes. Larger systems would be needed to sharpen this.

B. Protocol dependence

The Lyapunov gap is systematically larger under post-selection than under perfect correction in the DW-off case, but comparable or smaller in the DW-on case at the largest sizes. This reversal deserves attention.

Under post-selection the trajectory is deterministic: the circuit always selects the “no particle” outcome for all Wannier channels. This singles out a particular basin of attraction whose effective contraction rates can differ substantially from those sampled by the Born-rule ensemble. In the DW-off case the selected trajectory lies deep in the interior of the ergodic basin, so the gap is large. In the DW-on case the post-selected trajectory passes through configurations where the domain-wall charge fluctuates in a specific way, and the result-

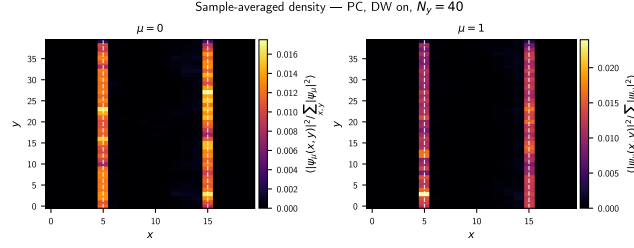


FIG. 8. Sample-averaged component-resolved density $\langle \rho_\mu(x, y) \rangle_s$ of the min- $|\lambda|$ eigenvector for $N_y = 40$, DW on, perfect correction. *Left*: orbital $\mu = 0$. *Right*: orbital $\mu = 1$. Each panel is independently normalised to unit total weight. White dashed lines mark the domain-wall columns $x = 5$ and $x = 15$. Both components are sharply localized at the wall interfaces and extend uniformly in y .

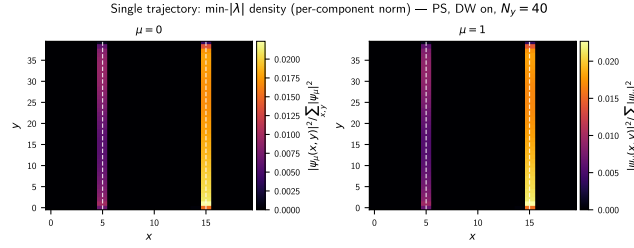


FIG. 9. Same as Fig. 8 but for the post-selection trajectory ($S = 1$). The left-right symmetry between the two walls is broken: one wall carries substantially more weight than the other, reflecting the asymmetry introduced by the deterministic outcome-0 protocol.

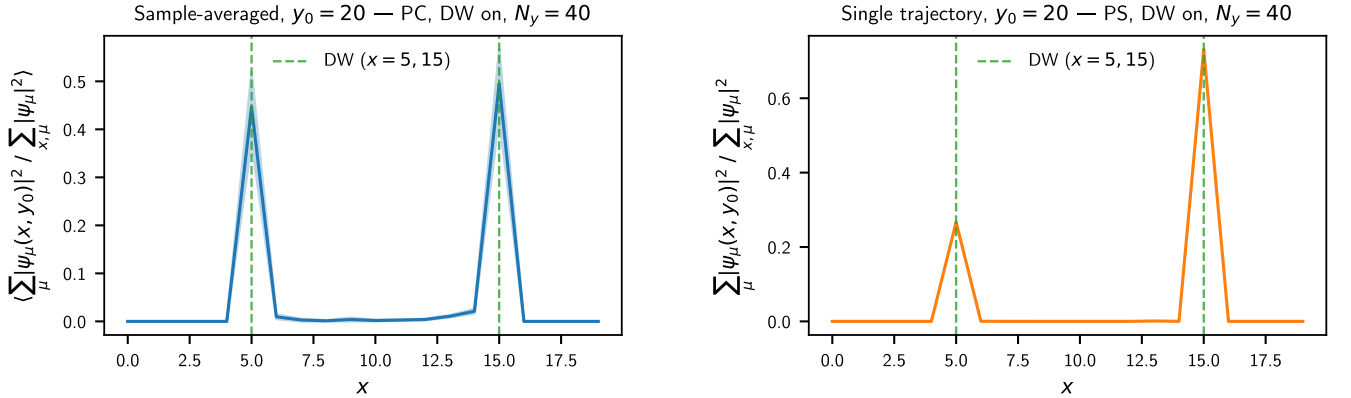


FIG. 10. Component-summed transverse profile $p(x|y_0)$ at $y_0 = N_y/2 = 20$ for $N_y = 40$, DW on. *Left*: perfect-correction sample average (± 1 s.e. band). *Right*: post-selection single trajectory. Green dashed lines mark $x = 5$ and $x = 15$. Both protocols show a profile peaked sharply at the wall columns with no weight in the trivial bulk; the PC average is approximately symmetric while the PS trajectory is asymmetric.

ing Lyapunov exponent can be even smaller than the perfect-correction average (e.g., $\Delta_{\lambda N_y=40}^{\text{PS}} = 0.0076$ vs $\Delta_{\lambda N_y=40}^{\text{PC}} = 0.024$). This suggests that the post-selected trajectory preferentially samples the critical manifold of the domain-wall sector.

The above observation reinforces the point made in Ref. [2] that perfect correction and post-selection are genuinely inequivalent ensembles: the physically meaningful observable is the Born-rule average over perfect-

correction trajectories, not the post-selected single trajectory.

C. Connection to the chiral domain-wall critical theory

The broader context established by the $N_x = 20$ campaign is that the adaptive circuit prepares a quasi-1D chiral critical ensemble on the domain wall, with entan-

glement entropy scaling approaching the chiral-fermion value $c_{\text{eff}} \approx 1$ and modular-charge velocities that change sign between the two walls. The Lyapunov analysis presented here adds an independent diagnostic:

1. The collapse of Δ_λ by more than an order of magnitude in the DW-on relative to the DW-off configuration confirms that the domain-wall sector introduces a slowly-decaying mode absent in the uniform bulk.
2. The monotonic decrease of Δ_λ with N_y is consistent with a mode that becomes truly marginal (gapless) as $N_y \rightarrow \infty$, as expected for a chiral CFT on a cylinder of circumference N_y .
3. The correlation between Δ_λ and the real-space Chern marker on a sample-by-sample basis locates the marginal mode in the topological boundary sector, not in the trivial bulk.
4. The neutral eigenvector (`lyapunov_min_abs_vector`) is a candidate for the primary field of the boundary CFT. Direct inspection of the component-resolved density $\rho_\mu(x, y)$ and the transverse profile $p(x|y_0)$ at $y_0 = N_y/2$ confirms that the weight is sharply concentrated at $x \in \{5, 15\}$ and extended uniformly in y , with negligible support in the trivial bulk. This is the expected signature of a chiral edge mode localized at the domain-wall interfaces.

D. Open questions

- *Functional form of the gap closing.* Is $\Delta_\lambda \sim N_y^{-1}$ (consistent with $c = 1$ Luttinger liquid / chiral fermion) or a different power? Sizes $N_y \geq 80$ with more samples are needed.
- *Spectrum near zero.* Is there a single near-zero mode or a band? The full sorted spectrum (Fig. 1) suggests only one mode approaches zero, but a systematic count as a function of N_y would test

whether the number of near-zero modes scales with the wall length.

- *N_y dependence of the cycle-time to convergence.* The gap does not plateau within $T = N_y$ cycles for the DW-on runs, suggesting the asymptotic Lyapunov regime has not been reached. Longer runs (e.g., $T = 5N_y$) are needed to extract the true long-time exponents.

V. CONCLUSION

We have computed finite-time Lyapunov spectra for the class-A adaptive fermionic circuit across three system sizes and four geometric/protocol configurations, using a full-rank QR Benettin algorithm propagated alongside the covariance-matrix dynamics. The results establish a sharp distinction between the ergodic (DW-off) and near-critical (DW-on) sectors. In the DW-off sector the Lyapunov gap is $\Delta_\lambda \sim 0.7\text{--}0.95$, large and size-independent, confirming rapid ergodic relaxation to a Chern-1 state. In the DW-on sector the gap collapses to $\Delta_\lambda \sim 0.007\text{--}0.027$, decreases monotonically with system size, and correlates on a sample-by-sample basis with deviations of the real-space Chern marker from integer quantization. These three signatures—gap suppression, size-driven closing, and Chern-marker correlation—collectively support the interpretation that the domain-wall circuit is proximate to a marginal fixed point associated with the chiral boundary CFT. The neutral tangent eigenvector has been shown to carry its weight sharply at the domain-wall interfaces ($x \in \{5, 15\}$), extended uniformly along the wall, constituting direct numerical evidence for the chiral boundary mode as the primary field of that theory.

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[1] A. Kitaev, “Anyons in an exactly solved model and beyond,” *Ann. Phys.* **321**, 2 (2006).
 [2] A. Bhuiyan, G.-Y. Pan, and C.-M. Jian, “Class-A fermionic adaptive circuits,” (2026), in preparation.

[3] G. Benettin, L. Galgani, A. Giorgilli, and J.-M. Strelcyn, “Lyapunov characteristic exponents for smooth dynamical systems and for Hamiltonian systems; a method for computing all of them,” *Meccanica* **15**, 9 (1980).